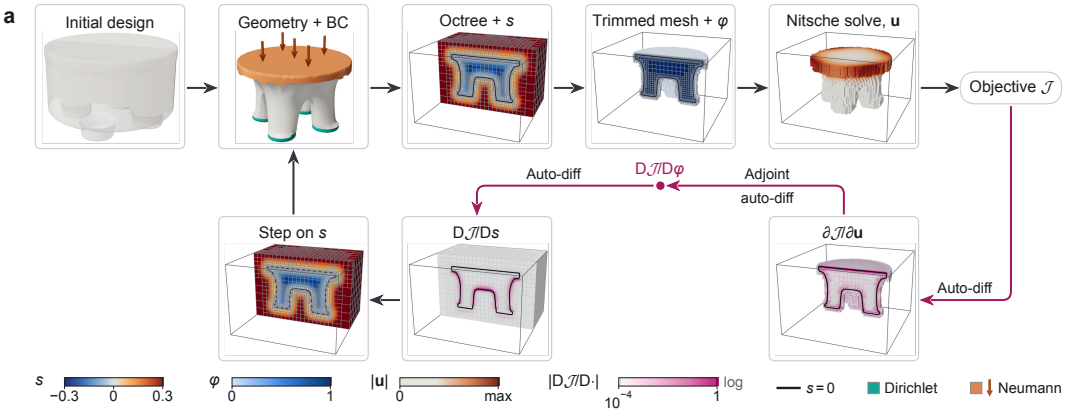

a
Initial design
Geometry + BC
Octree + s
Trimmed mesh + φ
Nitsche solve, u
Objective 𝒥
Auto-diff
D𝒥/Dφ
Adjoint
auto-diff
Step on s
D𝒥/Ds
∂𝒥/∂u
Auto-diff
s
−0.3
0
0.3
φ
0
1
|u|
0
max
|D𝒥/D·|
10−4
log
1
s = 0
Dirichlet
Neumann


b
Shape optimization
101 iterations, 4.0 min wall clock
Iteration 0
20
40
60
80
100